

VMS BMS Weekly Progress Reports

VMS Battery Management System Weekly Progress Reports

Report date: 13/01/2026 (DD/MM/YYYY)

- **Last Week (time in hours)**

- Team Review
 - Writing up TED talk summary to read to class (4)
- Jose Ramirez
 - Practice to read section of TED Talk (2)
- Hussein Khodor
 - Wrote up the TED Talk for the team since I was absent
- David Fransis
 - Read a little of the project manual, prepared for TED Talk
- Fangdong Yuan
 - Prepare for TED talk (1)

- **Next week**

- Team Plan
 - Revise Project Proposal Gantt Chart (2)
 - Expand on Verification Plan (2)
- Jose Ramirez
 - Expand on Verification Plan (2)
- Hussein Khodor
 - Get started on more in-depth research and broaden knowledge on batteries to kickstart developing a prototype
- David Fransis
 - Research Battery Systems (Analog Devices)
- Fangdong Yuan
 - Research for Battery Systems and chips

- **Blocked**

- Team Blocks
 - Waiting for faculty advisor to be available to set up a meeting to guide our research

- Jose Ramirez
 - N/A
- Hussein Khodor
 - N/A
- David Fransis
 - N/A
- Fangdong Yuan
 - N/A

Report date: 16/01/2026

• **Last Week**

- Team Review
 - Research into Analog devices (1)
 - Revise Project Proposal for Final Submission (2)
 - Write up Team Contract (3)
- Jose Ramirez
 - Set up collaboration site (2)
 - Research into Analog Devices BMS products (1)
 - Expand Verification Plan (1)
- Hussein Khodor
 - Finished up with preliminary research
- David Fransis
 - Revise Documentations
- Fangdong Yuan
 - Finish research

• **Next week**

- Team Plan
 - Battery Talk with Faculty Advisor (1)
 - Research further Analog Devices BMS products and their documentation (3)
- Jose Ramirez
 - Read into Analog Devices Secure Authentication and Communication devices BMS' (1.5)
- Hussein Khodor

- Conduct further in-depth research about battery management systems and try to connect that to a preliminary prototype
- David Fransis
 - More research on battery systems
- Fangdong Yuan
 - More research on Analog Devices BMS products
- **Blocked**
 - Team Blocks
 - No blocks so far
 - Jose Ramirez
 - No blocks so far
 - Hussein Khodor
 - No blocks so far
 - David Fransis
 - No blocks so far
 - Fangdong Yuan
 - No blocks so far

Report date: 23/01/2026

- **Last Week**
 - Team Review
 - Researched Analog devices
 - Updated Gantt Chart for Project Schedule and transferred it to GanttProject
 - Jose Ramirez
 - Set up “Meeting Notes” document (1)
 - Researched Analog Devices Secure Authenticator and Communication products (2)
 - Hussein Khodor
 - Conducted in-depth research about potential prototype components
 - David Fransis
 - Research into Cell and Pack Monitoring Kit AD-CELLPACKBM-SL (2)
 - Fangdong Yuan
 - research for BMS architecture, balance and some concepts

- **Next week**

- Team Plan
 - Everyone Thoroughly read FSAE Rule Book section EV and F.10—F.11
 - Right down which rules affect our design of Battery pack and management system and why
 - Determine if charging is done during competitions
 - Research Lithium battery Chemistry
 - Choose which Chemistry: Lithium-Nickel-Manganese-Cobalt or Lithium-Iron-Phosphate
 - Design for the cells, modules, battery pack
- Jose Ramirez
 - Research Battery Chemistries
- Hussein Khodor
 - Conduct in-depth research about FSAE rule book as well as battery chemistry and pack design
 - Finish transferring over our project schedule from our spreadsheet Gantt chart into the GanttProject software
- David Fransis
 - write about the FSAE rulebook
- Fangdong Yuan
 - research for FSAE rule, HV safety, isolated battery

- **Blocked**

- Team Blocks
 - Need a battery pack specs to start BMS Research and Design
- Jose Ramirez
 - Doesn't know Data Formatting needed for the VCU
- Hussein Khodor
 - N/A
- David Fransis
 - N/A
- Fangdong Yuan
 - N/A

Report date: 30/01/2026

• Last Week

- Team Review
 - Researched on Lithium chemistries
 - Learned Battery Safety and properties
 - Documented how rules affect our pack and BMS designs
- Jose Ramirez
 - Researched NMC Chemistry (2)
- Hussein Khodor
 - Researched LFP and NMC Chemisteis (2)
 - Editing gantt chart (2)
- David Fransis
 - Finished writing about the FSAE rulebook (2)
- Fangdong Yuan
 -

• Next week

- Team Plan
 - Fill out AHP matrices
 - Start research and development on Battery module designs
 - Look at old VMS EV battery from old vehicle
 - Learn CAN_FD Protocol for future development of BMS
- Jose Ramirez
 - Fill out Decision matrix for Battery Chemistry
 - Take CAN-FD courses
- Hussein Khodor
 - Research and explore possible battery pack designs to implement in a small-scale prototype
- David Fransis
 - Research battery pack designs / research old designs
- Fangdong Yuan
 - Research battery pack designs/ shape/ size

• Blocked

- Team Blocks
 - Same as last week
- Jose Ramirez

- Same as last week
- Hussein Khodor
 - N/A
- David Fransis
 - N/A
- Fangdong Yuan
 - N/A

Report date: 06/02/2026

• **Last Week**

- Team Review
 - Decided on LFP battery chemistry
 - Look into ICs in Open Source BMSs such as Libre Solar BMS
 - More research into the ICs of Open Source BMSs
- Jose Ramirez
 - Finished AHP for battery chemistry (3)
 - Decided for LFP chemistry
 - Wrote BMS L1 Block Diagram First Draft (1)
- Hussein Khodor
 - Explored open-source battery pack designs and BMS designs (2)
 - Looked at different BMS components that we could use for putting together our preliminary BMS prototype (1)
- David Fransis
 - Spoke with one of the ECU's members and asked him to join us on next Tuesday meeting, to discuss what chips they're using and how we can communicate. (0.5)
- Fangdong Yuan
 - Studied the architecture of the BMS (5)
 - Research the functions of different units in circuit (5)

• **Next week**

- Team Plan
 - Create list of ICs and evaluation boards to purchase
 - Make sure to get 2 pieces of each item, 4 if possible

- Email Andrew the completed list
- Update Block Diagram and start schematic for BMS modules
- Design simple 48V module for testing
 - Find a Battery welding station for EPL
- Create Unit Tests for new ICs
- Jose Ramirez
 - Update Block Diagram
 - Configure BMS modules
 - Create KiCAD schematic for a BMS module
- Hussein Khodor
 - Compile a list of possible BMS components that we can use to put together our preliminary BMS prototype
- David Fransis
 - look up a single model battery design
 - look up batteries soldering
- Fangdong Yuan
 - Research on Required Modules for BMS System
 - Reference to Industry Solutions
 - Comparative Study of Centralized, Distributed, and Daisy Chain Architectures
- **Blocked**
 - Team Blocks
 - We have cleared our blocks
 - Jose Ramirez
 - N/A
 - Hussein Khodor
 - N/A
 - David Fransis
 - N/A
 - Fangdong Yuan
 - N/A

Report date: 13/02/2026

• Last Week

- Team Review
 - Researched for Cell and Pack Monitor ICs from TI BQ79XXX
 - Read into using the NUCLO-C092RC eval board
 - Found Spot Battery Welder
- Jose Ramirez
 - Read STM32 User Manuals (2)
- Hussein Khodor
 - Researched a bunch about different cell and pack monitor ICs from TIs family of chips (5)
 - Brainstormed the initial design architecture of the preliminary BMS prototype with scaling in mind (5)
- David Fransis
 - Found a Battery welder (2)
- Fangdong Yuan
 - Compare the pros and cons of these three architectures in terms of cost, wiring complexity, sampling accuracy, and reliability (3)
 - Investigate reference designs from semiconductor giants like TI and AD (3)
 - Study for Battery Front-End (AFE), Battery Control Unit (BCU), Balancing Module (voltage consistency), Communication Module (CAN/RS485/Daisy Chain), and Safety Protection Module-Isolation circuit (4)

• Next week

- Team Plan
 - Fill out Decision Matrices for the following
 - Cell/Pack Monitoring IC
 - Battery Protection Circuitry
 - Learn CAN interface
 - Define messages for CAN Database
 - Send to VCU team
- Jose Ramirez
 - Learn CAN interface
 - Define CAN Messages
 - Further read into NUCLO-C092RC manuals and how to incorporate Zephyr
- Hussein Khodor

- Decide on a cell and pack monitor IC from TI's family of chips
- Decide on the initial design architecture of the preliminary BMS prototype
- Give Andrew a list of components that we need to begin our prototyping
- David Francis
 - Define messages for CAN Database
 - Send it and discuss it with the VCU team
- Fangdong Yuan
 - Decide on Confirmation of Daisy Chain as the Primary Architecture
 - In-depth Research on TI Chips--BQ796xx
 - Research on Circuit Isolation Methods for HV and LV Systems
- **Blocked**
 - Team Blocks
 - Determine Architecture to start designing Circuit Board
 - Need ICs and items for start firmware development
 - Jose Ramirez
 - Need Hardware and eval boards to test programs
 - Hussein Khodor
 -
 - David Francis
 - N/A
 - Fangdong Yuan
 -

Report date: 20/02/2026

- **Last Week**
 - Team Review
 - Sent our order for the ICs and components in our MVP and future development
 - Decided on which Monitoring IC to use
 - Learned more of technical details about CAN
 - Jose Ramirez
 - Learned CAN Protocol Basics (4)
 - Frame Types and Bit Time

- Hussein Khodor
 - Made and gave Andrew the list of components that we need to begin our prototyping (4)
 - Decided on the cell and pack monitor IC that we are using (2)
 - Decided on the initial design architecture of the preliminary BMS prototype (2)
- David Fransis
 - looked into Kicad Libraries for the chips we're planning on using (2)
 - found out about an alternative website that is similar to Kicad called 'EasyEDA' (1)
- Fangdong Yuan
 - Updated the list of what the VCU team should expect from us. (1)

• **Next week**

- Team Plan
 - Implement development environment for firmware development
 - Begin PCB design
 - Decide Protection Circuitry for battery cells/modules
- Jose Ramirez
 - Finish CAN Protocol Basics and CAN-FD network implementation
 - Learn to dual boot arch Linux with Windows and development environment on the former OS
- Hussein Khodor
 - Get started on interfacing with the BQ79656 evaluation module
 - Learn about how to program the NUCLEO-C092RC microcontroller to communicate with the BQ79656 evaluation module over SPI or UART
- David Fransis
 - Learn Kicad and look up EasyEDA
 - Design our own Kicad chips
- Fangdong Yuan
 - Update System Archeture Block diagram
 - Learn KiCAD

• **Blocked**

- Team Blocks
 - Need ICs and items for start firmware development
- Jose Ramirez
 - Need 511-NUCLEO-C092RC for development
- Hussein Khodor

- Need BQ79656EVM for development
- Need NUCLEO-C092RC for development
- David Fransis
 - Learning KiCAD
- Fangdong Yuan
 - Learning KiCAD

Report date: 27/02/2026

• **Last Week**

- Team Review
 - Updated Level 1 Block Diagram
 - Started working on KiCAD
 - Had to make IC schematics from scratch
 - Development Environments are ready
- Jose Ramirez
 - Setup zephyr development environment and built .elf for STM32 Dev board (3)
 - Learned some of the CAN error correction and synchronization setup (2)
- Hussein Khodor
 - Familiarized myself with the UART communication protocol (2)
 - Learned about how to interface with the NUCLEO STM32 family (1)
- David Fransis
 - Familiarized myself with KiCad
- Fangdong Yuan
 - Design the overall BMS architecture – signal + high-voltage + low-voltage systems (10)

• **Next week**

- Team Plan
 - design KiCAD 1.0 schematic
 - Learn rest of CAN-FD and look at current VCU CAN database
- Jose Ramirez
 - Learn remainder of CAN Basics and Implements CAN-FD Network
 - Look at cbd file from VCU team
- Hussein Khodor

- Continue to learn about the NUCLEO STM32 family and how to use it to communicate via UART to the BQ79656EVM
- David Fransis
 - Started a Kicad schematic draft
- Fangdong Yuan
 - Study the units required for each system
 - Design circuit diagram
- **Blocked**
 - Team Blocks
 - Hardware hasn't been delivered yet
 - Jose Ramirez
 - Need 511-NUCLEO-C092RC for development
 - Hussein Khodor
 - Need BQ79656EVM for development
 - Need NUCLEO-C092RC for development
 - David Fransis
 - Done with the first KiCad schematic draft
 - Fangdong Yuan
 - Separate LV/HV systems

Report date: 06/03/2026

- **Last Week**
 - Team Review
 - Ordered some of the parts for development
 - Jose Ramirez
 - Help write up TED Talk Slides (2)
 - Hussein Khodor
 - Contributed to TED Talk slides (1)
 - Set up development environment for the NUCLEO STM32 family using STM32CubeIDE and STM32CubeMX (4)
 - Interfaced with NUCLEO STM32 board and successfully flashed firmware onto it (2)

- Successfully communicated to my NUCLEO STM32 board from my PC using UART (3)
- David Fransis
 - Helped write up TED talk slides/prepared for it
- Fangdong Yuan
 - Updated BMS hardware architecture--HV/LV/SIGNAL (8)
 - TED talk slide (1)

• Next week

- Team Plan
 - Learn more of development environment
 - Hook up dev boards so Faculty Advisor can understand
 - Explain how Monitoring chip works
 - Learn CAN @Jose Ramirez!
 - Learn Isolation Protocols
 - Write Block Diagram V1.5 that clarifies the signals and module definitions
- Jose Ramirez
 - Learn CAN @Jose Ramirez!
 - Read about CAN Development and Database given by VCU Team
 - Read about Zephyr Kernel, Shell, and drivers
- Hussein Khodor
 - Read about Zephyr Kernel, Shell, and drivers
 - Migrate development environment over to Zephyr
- David Fransis
 - Learn more about HV-LV Isolation
- Fangdong Yuan
 - Finish BMS hardware architecture v1--HV/LV/SIGNAL

• Blocked

- Team Blocks
 - Lack of Hardware for Development
- Jose Ramirez
 - No Dev Boards for Firmware
- Hussein Khodor
 - No Dev Boards for Firmware
- David Fransis

- N/A
- Fangdong Yuan
 - N/A

Report date: 14/03/2026

• Last Week

- Team Review
 - Determined how monitoring chip bus should be assembled
 - Not much Software Development
- Jose Ramirez
 - Read about the Zephyr Peripherals (1)
- Hussein Khodor
 - Read about Zephyr Kernel, Shell, and drivers (2)
- David Fransis
 - learned more about HV-LV isolation
 - looked into isolation devices that could work for our HV-LV
- Fangdong Yuan
 - updated BMS diagram v2--HV/LV/SIGNAL(5)

• Next week

- Team Plan
 - Determine if the we want the pure communication IC eval board
 - Prepare for Firmware Development when components arrive
 - Design HV-LV Isolation for schematic and PCB
- Jose Ramirez
 - Learn CAN @Jose Ramirez!
 - Read about CAN Development and Database given by VCU Team
 - Read about Zephyr Kernel, Shell, Hardware Support, and drivers
- Hussein Khodor
 - Migrate development environment over to Zephyr
- David Fransis

- Make a list of the isolation devices in consideration
- Update KiCad
- Fangdong Yuan
 - finished BMS diagram v2--HV/LV/SIGNAL

• **Blocked**

- Team Blocks
 - Still waiting for Hardware
 - Finals next week
- Jose Ramirez
 - Waiting for STM32 eval board
- Hussein Khodor
 - Waiting for STM32 eval board
- David Fransis
 - Understanding the pinouts we'll be using
- Fangdong Yuan
 - n/a

Report date: 20/03/2026

• **Last Week**

- Team Review
 - Busy with finals
 - Block Diagram 2.0V complete
- Jose Ramirez
 - Was studying for finals and finishing course work (0)
- Hussein Khodor
 - Studying for and finishing up finals (0)
- David Fransis
 - Was studying for finals and finishing course work (0)
- Fangdong Yuan
 - finished the diagram-v2(2)

• **Next week**

- Team Plan

- Add separate Current Sensor enclosing SHUNT resistor and battery charger for LV battery
- Program Loan STM32 eval board
- Jose Ramirez
 - Learn CAN [@Jose Ramirez](#) by Tuesday!
 - Read about CAN Development and Database given by VCU Team by Tuesday
 - Read about Zephyr Kernel, Shell, Hardware Support, and drivers by Tuesday
 - Program STM32 eval board
- Hussein Khodor
 - Migrate development environment over to Zephyr
- David Fransis
 - learn more about shunt and current measuring
 - Kicad schematics for BQ79616, BMU, and Shunt sense.
- Fangdong Yuan
 - Completed the overall system architecture design and clarified the functional blocks and key interface relationships.
- **Blocked**
 - Team Blocks
 - ICs order got canceled, ordered new set of ICs
 - Jose Ramirez
 - N/A
 - Hussein Khodor
 - Missing STM32 eval board
 - David Fransis
 - N/A
 - Fangdong Yuan
 - n/a

Report date: 27/03/2026

- **Last Week**
 - Team Review
 - Caught up with hardware design
 - Prepared for board design of monitoring and SHUNT modules

- Planning firmware development and finalizing CAN-FD specifications
- Jose Ramirez
 - Finished learn CAN/CAN-FD basics (3)
 - Wrote up most of the CAN-FD specifications concerning the BMS (3)
 - Set course for Firmware development (1)
- Hussein Khodor
 - Finished migrating development environment to Zephyr (2)
 - Successfully connected to and woke up the BQ79600EVM with my STM32 Nucleo C092RC board (3)
- David Fransis
 - Started Kicad files for Battery module, and the bridge communication
- Fangdong Yuan
 - On vacation
- **Next week**
 - Team Plan
 - System Architecture schematics will be done by Friday **before meeting**
 - Firmware Deliverables
 - UART communication done and perhaps read voltage from monitoring chip by Monday
 - CAN shell set up and able to send BMS messages by Monday
 - Jose Ramirez
 - Configure CAN-FD controller, set up shell, and send CAN message example from database using can command
 - Read Rulebook and write up CAN-FD BMS Sequence Message Specifications
 - Write up the first half of Final Report
 - Hussein Khodor
 - Communicate with BQ79600EVM over UART using Zephyr UART API and polling
 - David Fransis
 - Get the battery module schematic board done
 - Get the bridge communication schematic board done
 - Fangdong Yuan
 - KiCad schematic design
- **Blocked**
 - Team Blocks

- One team member is on vacation
- Jose Ramirez
 - N/A
- Hussein Khodor
 - N/A
- David Fransis
 - Missing Teammate for hardware development
- Fangdong Yuan
 - Out on vacation for spring break

Report date: 03/04/2026

• **Last Week**

- Team Review
 - Finished the BMU and Monitoring Schematics
 - Shell set up for UART and CAN
- Jose Ramirez
 - Accessed CAN Controller through Zephyr Shell (3)
- Hussein Khodor
 - Successfully communicated with BQ79600EVM over UART using Zephyr UART API and polling (5)
- David Fransis
 - done with the battery module, and the bridge communication schematics drafts,
- Fangdong Yuan
 - Become familiar with the KiCad schematic design workflow and project organization for subsequent development (8)

• **Next week**

- Team Plan
 - Hardware Development
 - @David Fransis SHUNT Current module schematic by Wednesday
 - SHUNT resistor
 - Current Sensor determined by Monday/Tuesday

- @Fangdong Yuan BMU revisions (adding the STM32 and CAN Bus electronics) by Wednesday
 - @Jose Ramirez will report what CAN Electronics are need for the BMU to communicate to bus.
 - STM32C092RCT6 IO Ports for BMS
 - USART2TX: PA2/Pin_19
 - USART2RX: PA3/Pin_20
 - FDCAN_TX: PD1/Pin_51
 - FDCAN_RX: PD0/Pin_50
 - FDCAN_STBY: PD2/Pin_52
 - Get start on PCB development
 - Hopeful drafted by end of week
 - Firmware Development
 - @Hussein Khodor Implement communicating with the BQ79600EVM from the STM32 Nucleo C092RC using Zephyr UART API and asynchronous UART.
 - Connect and communicate with the BQ79616EVM board to read battery cell data using our STM32 Nucleo C092RC connected to our BQ79600EVM.
 - @Jose Ramirez Configure CAN Controller and verify that Monitoring Message is received by mock VCU by Monday
 - Prototype integration start on Wednesday
 - Structures and shell commands for storing voltage [Module:Cell] readings and sending and receiving them on them on CAN Bus.
 - Hopefully start Code for Firmware by end of week
 - Jose Ramirez
 - Write Configuration File for CAN Controller, both MCU for BMS and mock VCU, by Monday
 - Send Montioring Messages
 - Integrate with UART communication to read voltage readings
 - Hussein Khodor
 - Communicate with BQ79600EVM over UART using Zephyr UART API and asynchronous DMA
 - David Fransis
 - Finding a current sensor device
 - work on making a SHUNT Current module schematic
 - Fangdong Yuan

- Split the overall schematic into multiple functional pages to establish a clearer hierarchical structure.

- **Blocked**

- Team Blocks
 - N/A
- Jose Ramirez
 - Large Course load this term
- Hussein Khodor
 - N/A
- David Fransis
 - N/A
- Fangdong Yuan
 -

Report date: 10/04/2026

- **Last Week**

- Team Review
 - Hardware
 - Found series of current sensor modules
 - Set up for BMU schematic revisions
 - Firmware
 - Complete Asyn interface to Monitoring module
 - CAN structures for sending to VCU
- Jose Ramirez
 - Found that isolation faults are monitored by the IMD (1)
 - BMS will only transmit to the CAN Bus (1)
- Hussein Khodor
 - Successfully communicated with BQ79600EVM over UART using Zephyr UART API and asynchronous DMA (3)
- David Fransis
 - Found a current sensor (turned out not good enough)

- Fangdong Yuan
 - Developed the initial Power Page and optimized the power supply path and supporting components. (10)

• **Next week**

- Team Plan
 - Hardware
 - Current Monitoring Module
 - Use current sensor of ADC potentially connected
 - BMU schematic
 - Fill in other sub sheets
 - Firmware
 - Create mock VCU Module
 - Read Voltage Values
 - Transmit Message of voltage data
- Jose Ramirez
 - Write up API for Data calculation and CAN transmission
 - Write up Final Report Draft
 - Include Test Plan
- Hussein Khodor
 - Communicate with BQ79616EVM via daisy-chain from BQ79600EVM to read all cell voltages from 6S prototype battery pack
- David Fransis
 - look up current sensors that could handle 400+ Amps (With I²C preferred).
 - After finding the current sensor, make a current monitor schematic.
- Fangdong Yuan
 - Built the BQ79600 page and defined its power, communication, and interface connections.

• **Blocked**

- Team Blocks
 - N/A
- Jose Ramirez
 - Large Course load this term
- Hussein Khodor
 - N/A

- David Fransis
 - N/A
- Fangdong Yuan
 - kicad workshop

Report date: 17/04/2026

• Last Week

- Team Review
 - Hardware
 - Current Sensor Module
 - SHUNT IC chosen
 - Schematic near completion
 - BMU revisions
 -
 - Software
 - Voltage Monitoring has been established
 - Code Modularized
- Jose Ramirez
 - Not Much (0)
- Hussein Khodor
 - Successfully communicated with BQ79616EVM via daisy-chain from BQ79600EVM to read all cell voltages from 6S prototype battery pack (6)
- David Fransis
 - Found a current sensors that could handle 400+ Amps (With I^2C)
 - Started a a current monitor schematic
- Fangdong Yuan
 - Continued the design of the CAN and MCU pages to improve the main control and communication framework. (8)
 - Built the real circuit (3)

• Next week

- Team Plan
 - Hardware

- Finish up both schematics
 - BMU Power Supply sheet: Fix protection electronics
 - Group the STM32 pins by designation (PA, PB, PD, ...)
- Start PCB design
- Decide Cell connector
 - Most likely Locking polarized automotive connector
- Software
 - Create L1 Block Diagram
 - Create API for firmware
 - **Write Final Report Draft** @Jose Ramirez
- Jose Ramirez
 - Write Software Block Diagram
 - Set up Firmware API
 - **Write up Final Report Draft**
 - **Include Test Plan**
- Hussein Khodor
 - Communicate with BQ79616EVM via daisy-chain from BQ79600EVM to read all on-board thermistor temperatures
- David Fransis
 - Finish the current monitor schematic
 - Do a write up about the schematic and the sensor we chose.
 - Start designing the PCBs
- Fangdong Yuan
 - Started preparing footprints and coordinating the interfaces with the shunt and bq79616 modules, while continuously refining the overall design.
- **Blocked**
 - Team Blocks
 - N/A
 - Jose Ramirez
 - Large Course load this term
 - Hussein Khodor
 - N/A
 - David Fransis

- N/A
- Fangdong Yuan
-

Report date: 24/04/2026

• **Last Week**

- Team Review
 - Hardware
 - Schematics Development mostly finished
 - Software
 - Temperature Sensing added
- Jose Ramirez
 - Wrote on Software block Diagram (2)
- Hussein Khodor
 - Successfully communicated with BQ79616EVM via daisy-chain from BQ79600EVM to read all on-board thermistor temperatures (5)
- David Fransis
 - Finished the current monitor schematic
 - Did a write up about the schematic and the sensor we chose.
- Fangdong Yuan
 - Refined the BMU schematic structure, including power, MCU, CAN, and BQ79600 sections.(9)
 - Reviewed BMU power rails, connector definitions, signal naming, and key component footprints. (5)

• **Next week**

- Team Plan
 - Hardware
 - Finish Revisions
 - Get PCB v1.0 done by Monday
 - Software
 - Configure the Voltage and Temperature Thresholds
 - Create CAN-FD Transmission
- Jose Ramirez

- Set up Firmware API
 - Create Test CAN-FD node
- **Write up Final Report Draft**
 - **Include Test Plan**
- Hussein Khodor
 - Configure and read fault status on the BQ79600EVM and BQ79616EVM devices
- David Fransis
 - Update the Current monitor schematic
 - Finish the PCB layout for the Current monitor
- Fangdong Yuan
 - Improved BMU interface organization for debugging, communication, and external connections.
 - Started BMU PCB layout preparation, including component placement and basic routing planning.
- **Blocked**
 - Team Blocks
 - Need CAN-to-USB converter
 - Jose Ramirez
 - Large Course load this term
 - Hussein Khodor
 - N/A
 - David Fransis
 - N/A
 - Fangdong Yuan
 -

Report date: 01/05/2026

- **Last Week**
 - Team Review
 - Hardware
 - BMU and SHUNT Sensing Modules v1.0 layout nearing completion
 - Monitoring module not started yet
 - Software

- Has been staggent
- Jose Ramirez
 - Wrote up Test Plan First Draft (2)
- Hussein Khodor
 - Successfully configured and read fault status on the BQ79600EVM and BQ79616EVM devices (5)
- David Fransis
 - Updated the Current monitor schematic
- Fangdong Yuan
 - Completed the main BMU PCB layout, including placement, routing, grounding, and power distribution.(10)
 - Organized the BMU 4-layer board structure with signal layers, ground plane, and power zones. (8)
- **Next week**
 - Team Plan
 - Hardware
 - Add test points using Test_Point_Square
 - Assign footprint with 1.5 mm diameter with 0.7 mm drill
 - Add LED circuitry to following:
 - BMU: Power 5.0V and 3.3V
 - SHUNT Sensing: Power, Iso Power, and Post-protection ALERT signal
 - Monitoring Module: Power
 - Start Monitoring Module layout and complete it by Monday
 - Software
 - Research into Zephyr OS interrupts
 - Port the BMU Board to a repo board directory as shown in [the Zephyr Manual](#)
 - @Jose Ramirez Finish CAN transmission
 - Start doing interrupt handling for faults on and off board.
 - Demonstration and Poster
 - Mention in background that the existing design was scraped 7 years ago
 - Show schematics and layouts
 - Jose Ramirez
 - Continue working on Final Report

- @Jose Ramirez Finish CAN transmission
- Update firmware block diagram to v2.0
- Implement calculations and message formatting
- Start creating board directory for BMU
- Hussein Khodor
 - Configure cell balancing on the BQ79616EVM device
- David Fransis
 - Start the PCB design for the Current monitor
- Fangdong Yuan
 - Added BMU test points for power rails, communication signals, and debugging interfaces.
 - Ran ERC/DRC checks, fixed layout warnings, reviewed footprints, and pushed the updated BMU files to GitHub.
 - Focused on CMU schematic and layout design based on the BQ79616 cell monitoring module.
 - Reviewed CMU cell input labels, connector definitions, COM daisy-chain interface, and related footprints.
- **Blocked**
 - Team Blocks
 - Midterms this week and next
 - Jose Ramirez
 - Large Course load this term
 - Backlog of work from other classes
 - Hussein Khodor
 - ECE 586 midterm on Tuesday of next week
 - David Fransis
 - Work on Weekends
 - Fangdong Yuan
 - N/A

Report date: 08/05/2026

- **Last Week**
 - Team Review
 - Hardware

- Monitoring Module, now called Cell Monitoring unit (CMU), PCB first draft finished
 - Reorganization of the BMU schematic
- Software
 - Fault configuration completed
- Jose Ramirez
 - Busy catching up with Course work (0)
- Hussein Khodor
 - Successfully configured cell balancing on the BQ79616EVM device (3)
- David Fransis
 - Current monitor PCB draft done
- Fangdong Yuan
 - Improved CMU cell sense routing, grounding strategy, test points, and silkscreen labels.(16)
 - Checked CMU schematic-to-layout consistency and pushed the updated CMU files to GitHub for team review.
- **Next week**
 - Team Plan
 - Hardware
 - Removing unnecessary files from BMU libraries
 - Reviewing Checklists for remaining Modules
 - Software
 - Cell Balancing Started
 - CAN-FD Test Program should be done this week (Transmission only)
 - Start on Protection Interrupt system and Measurement Calculations/Packing
 - Jose Ramirez
 - Continue working on Final Report
 - @Jose Ramirez Finish CAN transmission
 - Update firmware block diagram to v2.0
 - Implement calculations and message formatting
 - Start creating board directory for BMU
 - Hussein Khodor
 - Communicate with and measure current from the INA238EVM device with current shunt
 - David Fransis
 - Check my PCB design with Andrew

- Fangdong Yuan
 - Discussed the BMU and CMU design approach with Andrew, including routing strategy, grounding, connector choices, and component selection.

- **Blocked**

- Team Blocks
 - N/A
- Jose Ramirez
 - Large Course load this term
- Hussein Khodor
 - N/A
- David Fransis
 - Andrew's availability
- Fangdong Yuan
 -

Report date: 15/05/2026

- **Last Week**

- Team Review
 - Hardware
 - Boards aren't order yet
 - no Faculty Advisor Approval
 - Most DRC and ERC issues fixed
 - Software
 - Protection Protection
 - CAN Test message Completed
- Jose Ramirez
 - Wrote CAN Test Message (4)
- Hussein Khodor
 - Successfully communicated with and measured current from the INA238EVM device with current shunt (2)
- David Fransis
 - Made small changes on the PCB design for the Current monitor (2)

- Fangdong Yuan
 - Prepared the updated KiCad files, BOMs, and design package for team review before final PCB ordering.(10)

• **Next week**

- Team Plan
 - Hardware
 - Fix any remaining PCB errors
 - Get Faculty Advisor Approval and Order boards
 - Work on Final Report Appendices
 - Developer's Manual
 - Bottom up test plan for subsystems
 - Software
 - Finish Cell Balancing
 - Work on SHUNT module moitoring
 - Set up interrupt handler for faults
 - Set up CAN messages
- Jose Ramirez
 - Continue working on Final Report
 - Update firmware block diagram to v2.0
 - Implement calculations and message formatting
 - Start creating board directory for BMU
- Hussein Khodor
 - Set up and configure overcurrent detection on the INA238EVM device with current shunt
- David Fransis
 - Fix any errors I have left on the circuit monitor design
- Fangdong Yuan
 - Reviewed and reorganized the BMU and CMU BOMs, including resistors, capacitors, ICs, connectors, switches, protection components, and power-related parts.(8)

• **Blocked**

- Team Blocks
 - PCBs not ordered yet
- Jose Ramirez
 - Large Course load this term

- Hussein Khodor
 - N/A
- David Fransis
 - N/A
- Fangdong Yuan
 -

Report date: 22/05/2026

• Last Week

- Team Review
 - Hardware
 - PCB design is complete
 - Waiting for Faculty Advisor Approval
 - Switching
 - Software
 - New CANmodule
 - Monitoring Program Finished
- Jose Ramirez
 - Added CANmodule (2)
- Hussein Khodor
 - Successfully set up and configured overcurrent detection on the INA238EVM device with current shunt (3)
- David Fransis
 - Fixed all errors for the Current Monitor circuit design (2)
 - Made BOM for the Current Monitor (3)
- Fangdong Yuan
 - several BOM items needed to be replaced with more suitable, common, or easier-to-source components.(12)

• Next week

- Team Plan
 - Hardware
 - Meet with VCU team for Prototype integration

- Keep nagging Advisor about PCB design review and ordering the boards
- Software
 - Finish CAN Message packing and CAN Integration with BMS monitoring
- Documentation
 - Write Final Report Draft
 - Write Ethics Design Evaluation Document
- Jose Ramirez
 - Write up Final Report Draft
 - Write Ethics Design Evaluation Document
- Hussein Khodor
 - Set up and configure the CAN communication to send data to the ECU team over CAN
- David Fransis
 - Figure out a way to organize and minimize our wires, so that we could attach them to the VCU car during demo without any problems.
 - Meet with VCU team for Prototype integration
- Fangdong Yuan
 - Optimized the CMU design, including cell sense routing, connector labeling, grounding strategy, test points, and silkscreen readability.
- **Blocked**
 - Team Blocks
 - Unable to order Boards in time
 - Switching to demoing with eval boards
 - Jose Ramirez
 - Large Course load this term
 - Hussein Khodor
 - N/A
 - David Fransis
 - N/A
 - Fangdong Yuan
 -

Report date: 29/05/2026

• Last Week

- Team Review
 - Technical Work
 - PCB development finished, Boards will be ordered on Saturday
 - All Firmware development complete
 - Only branch to CAN 2.0 change for Prototype Integration with VCU
 - Documentation
 - More of Final Report written up
 - More minor documentation done
- Jose Ramirez
 - Writing Final Report (1)
- Hussein Khodor
 - Successfully set up and configured CAN communication to send data to the ECU team over CAN (6)
- David Fransis
 - Finished The BOM for the Current Monitor (2)
 - Met with the VCU team (2)
- Fangdong Yuan
 - Checked schematic-to-layout consistency for both BMU and CMU after the BOM and layout updates to reduce mismatch or ordering risks.(8)

• Next week

- Team Plan
 - Technical Work
 - Mostly complete
 - Assembly soldering of Boards
 - Execute test plan to see if they're functional
 - Potential integration with demo
 - **Only** if boards and components arrive within the next week
 - Assemble extra battery pack for demonstrating stackable battery modules
 - Modify CAN firmware to transfer from CAN-FD to CAN 2.0
 - VulCANs **can't interpret CAN-FD**
 - Branch from main
 - Name it "Demo CAN"

- Documentation
 - Further expand on the test plan
 - Add more subsystem testing!
 - Do any revisions suggested by VMS and Andrew
 - Start on Poster
- Jose Ramirez
 - Further expand on the test plan and Final report
 - Organize google drive and find way to transfer ownership to VMS
 - Plan on moving WPR document to Drive
 - Help solder boards **if** they come within this next week
 - Execute top-down and bottom test plans to ensure functionality
 - Start on Poster
- Hussein Khodor
 - Clean up wiring mess on breadboard prototype setup
- David Fransis
 - Stay alert for any updates on the PCBs, so that if anything needed to be fixed I can fix it ASAP
 - Update PCBs documentations
- Fangdong Yuan
 - Check CMU+BMU+SHUNT Board
- **Blocked**
 - Team Blocks
 - N/A
 - Jose Ramirez
 - Large Course load this term
 - Hussein Khodor
 - N/A
 - David Fransis
 -
 - Fangdong Yuan
 -

Report date: 05/06/2026

- **Last Week**

- Team Review
 - Documentation
 - Most content of the final report is done
 - Appendices need to be done
 - Results, Postmortem, and Conclusion need to be written up
 - Poster barely started and is due on Sunday
 - Technical Work
 - Mostly finished rewiring
 - Only rewriting of CAN-FD to CAN 2.0 for demo integration with ECU
 - Boards Arrive between Tuesday and Wednesday of next week
 - Likely wouldn't be able to integrate with demo :-(
- Jose Ramirez
 - Written up First few section of on the Final Report in LaTeX(2)
 - Started the Capstone Poster (2)
- Hussein Khodor
 - Cleaned up wiring mess on breadboard prototype setup (15)
- David Fransis
 -
- Fangdong Yuan
 - Wrote Project Final Report
- **Next week**
 - Team Plan
 - Documentation
 - Finish last sections within the next **24 hours!**
 - Use the final report for the poster bullet points
 - Add the system architecture and prototype pictures to poster
 - Finish poster by **Sunday Morning!**
 - Technical work
 - Finish CAN 2.0 conversion (remember to have it as a separate branch to main)
 - If boards arrive on Tuesday or Wednesday, assemble and use project documentation to assemble, program, and test
 - Jose Ramirez
 - Finish Poster

- Help assemble boards
- Hussein Khodor
 - Add classic CAN (CAN 2.0) compatibility to code for integration with ECU team
 - Integrate and test BMS prototype with ECU prototype (hopefully with custom PCBs as well)
 - Expose XT60 female connector from battery to connect to ECU system
 - Help assemble custom PCB boards
- David Fransis
 - Work on the Project Final Report
 - Help assemble custom PCB boards
- Fangdong Yuan
 - Finish Project Final Report
 - Finish Project Poster
 - Solder CMU+BMU+SHUNT Board
- **Blocked**
 - Team Blocks
 - N/A
 - Jose Ramirez
 - Large Course load this term
 - Hussein Khodor
 - Final Exam preparation and studying
 - David Fransis
 - N/A
 - Fangdong Yuan
 -

Report date: 11/06/2026 (Project End!)

- **Last Week**
 - Team Review
 - Poster Finished Sunday Night
 - PCBs come in very late
 - All parts and boards arrived on Thursday
 - Managed to solder all the boards

- Main Final Report Complete!
- Prototype Integration with ECU was a success 🥳
- Flat system assembled!
- Jose Ramirez
 - Finished the Poster (3)
 - Written up Conclusion and Project Resources on Final Report (2)
 - Helped Soldering the PCBs (3)
- Hussein Khodor
 - Added classic CAN (CAN 2.0) compatibility to code for integration with ECU team (4)
 - Assembled Prototype into Flat System (2)
 - Integrated and tested BMS prototype with ECU prototype (sadly not with PCBs) (1)
 - Helped assemble custom PCB boards (4)
 - Exposed XT60 female connector from battery to connect to ECU system (1)
- David Fransis
 - Finished some sections on the Project Final Report
 - helped soldering CMU Board
- Fangdong Yuan
 - Finished Project Final Report(3)
 - Finished Project Poster(3)
 - Soldered CMU+BMU+SHUNT Board(12)